

Qhawe Ngcongo
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PROFESSIONAL SUMMARY

Computer Science graduate with hands-on experience designing and deploying production-ready mobile and backend systems. Skilled in Android (Kotlin), RESTful API development, cloud deployment, and relational database architecture. Experienced in integrating payment gateways, implementing offline-first mobile architectures, and building scalable backend services. Currently pursuing postgraduate studies in Data Analytics to strengthen data-driven system design and analytical capabilities. Seeking junior backend or software engineering roles.

Skills and Competencies

- Genuine eagerness to learn and adjust within a technical setting
- Strong critical thinking and ability to resolve problems
- Effective communication and strong people skills
- A collaborative mindset combined with professionalism
- Capable of following directions, completing tasks on time, and working with guidance
- Deep curiosity and passion for engineering, infrastructure, and mobility systems

TECHNICAL SKILLS

- Programming Languages
Kotlin, C#, JavaScript, SQL
- Mobile Development
Android (MVVM), Room, Firebase Realtime Database, Material Design
- Backend Development
Node.js, Express.js, [ASP.NET](#) Core, RESTful API Design, API Integration, JWT Authentication, Role-Based Access Control
- Databases
SQL Server, MySQL, SQLite (Room), PostgreSQL, Database Design, Query Optimization
- AI & OCR
Google Cloud Vision API, Document Text Extraction, Pattern Matching

- Cloud & DevOps
Render (Cloud Deployment), Vercel, Neon (Serverless PostgreSQL), GitHub Actions (CI/CD), Firebase
- Tools
Git, GitHub, Android Studio, Visual Studio, VS Code, Postman

EDUCATIONAL BACKGROUND

- **Highest standard passed: Matric**
Year: 2022
- Courses passed: Bachelor of Computer and Information Science in Application Development
Institution: Varsity College/Emeris
Year passed: 2025

Modules completed:
Data Structures & Algorithms | Database Systems | Software Engineering | Cloud Development | Web Development | Artificial Intelligence
- Current Studies:
- Postgraduate Diploma in Data Analytics (In Progress)
Institution: Varsity College / Emeris — Expected 2026

PROJECT/WORK EXPERIENCE

Document Management System (Full-Stack Web Application)

Technologies: React, Node.js, Express, PostgreSQL, Google Cloud Vision API, JWT, Render, Vercel

- Architected and deployed a production-ready document management system with secure JWT authentication and role-based access control (Admin, Approver, Viewer).
- Integrated Google Cloud Vision API for AI-powered extraction of vendor names, dates, amounts, VAT, and invoice numbers from uploaded documents.
- Implemented a complete 3-step approval workflow (Reviewer → Manager → Finance/Admin) with real-time status tracking and approval history.
- Built duplicate detection logic to prevent duplicate invoice uploads using multi-tier validation.
- Developed a comprehensive reporting module with spend summaries, vendor analysis, and tax/VAT reports with PDF/Excel export capabilities.
- Designed responsive frontend using React with custom CSS and deployed to Vercel.

- Engineered scalable PostgreSQL database schema with Neon (serverless PostgreSQL) and deployed backend to Render.
- Live Demo: <https://document-management-system-vercel.vercel.app>
- GitHub: github.com/Qhaweisthename/Document-management-system

Highway Hospice Management System (WIL Project)

Client: *Highway Hospice (NGO)*

Technologies: *Android (Kotlin), REST APIs, Firebase, PayFast API*

- Collaborated in a development team to build a production-ready mobile and web solution for a non-profit organization.
- Led Android frontend development, designing and implementing complete UI/UX using modern Android architecture.
- Integrated RESTful APIs to enable seamless communication between mobile and backend services.
- Implemented secure online donation processing using the PayFast payment gateway API.
- Utilized Firebase for authentication and real-time cloud data management.
- Contributed to database schema design and backend integration.
- Delivered a deployed application used by a real-world NGO.
- Google Play Link:
<https://play.google.com/store/apps/details?id=com.TheHighwayHospice.thehighwayhospiceapp>

Habit Tracking Application (Full-Stack Mobile System)

Technologies: *Android, Custom REST API, SQL Database, Render*

- Architected and developed a full-stack habit tracking system with a custom backend API.
- Designed and implemented RESTful endpoints for user and task management.
- Designed scalable relational database schema.
- Implemented offline-first functionality with local storage and data synchronization.
- Deployed backend services to cloud infrastructure using Render.
- Developed gamification logic to enhance user engagement and retention.
- GitHub: <https://github.com/Qhaweisthename/Habitflow-buildAPK>

BudgetWithUs – Full-Stack Android Finance Application

Technologies: *Kotlin, MVVM, Room, Firebase Realtime Database, GitHub Actions*

- Designed and developed a scalable personal finance management application.
- Implemented hybrid architecture combining local persistence (Room) with cloud synchronization (Firebase).
- Designed relational database schema for expense tracking and budget goals.
- Developed real-time budget monitoring logic with dynamic feedback mechanisms.
- Implemented CI/CD pipeline using GitHub Actions for automated builds.
- Applied Material Design principles for improved usability and accessibility.

- GitHub: <https://github.com/EmerisWVL/prog7313-poe-Qhaweisthename>

KEY STRENGTHS

- Backend system architecture & REST API development
- Database modeling & query optimization
- AI/OCR integration for document processing
- Role-based access control & authentication systems
- Offline-first mobile architecture
- Payment gateway integration
- Strong analytical & problem-solving skills
- Clean, structured, maintainable code

REFERENCES

Name: Zanele Mnguni

Position: Manager

Contact number: 073 534 8563